Bursar Whitepaper

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Keeps your books. *Never* breaks your rules.

Abstract

Bursar is an AI agent that runs a small business's money in USDC on Arc. It sends invoices, chases late payers, matches payments, pays approved bills, keeps a reserve, saves, and checks every wallet before money moves.

The agent decides, but the contract has the final say. Every business keeps its money in its own vault contract on Arc, and the owner's rules live inside it. The agent can only act within those rules, even if it gets something wrong or is tricked.

What the name stands for. A bursar is the person who keeps an institution's money: collecting fees, paying bills and keeping the books. The word comes from the Latin *bursa*, meaning purse. Bursar does that job for a small business, and it keeps the purse strings in the owner's hands.

The problem

Small businesses that get paid in USDC still do all of their money work by hand. An owner sends invoices from chat, checks a wallet to see who paid, matches each payment to an order from memory, sends awkward reminders, and pays suppliers one transfer at a time.

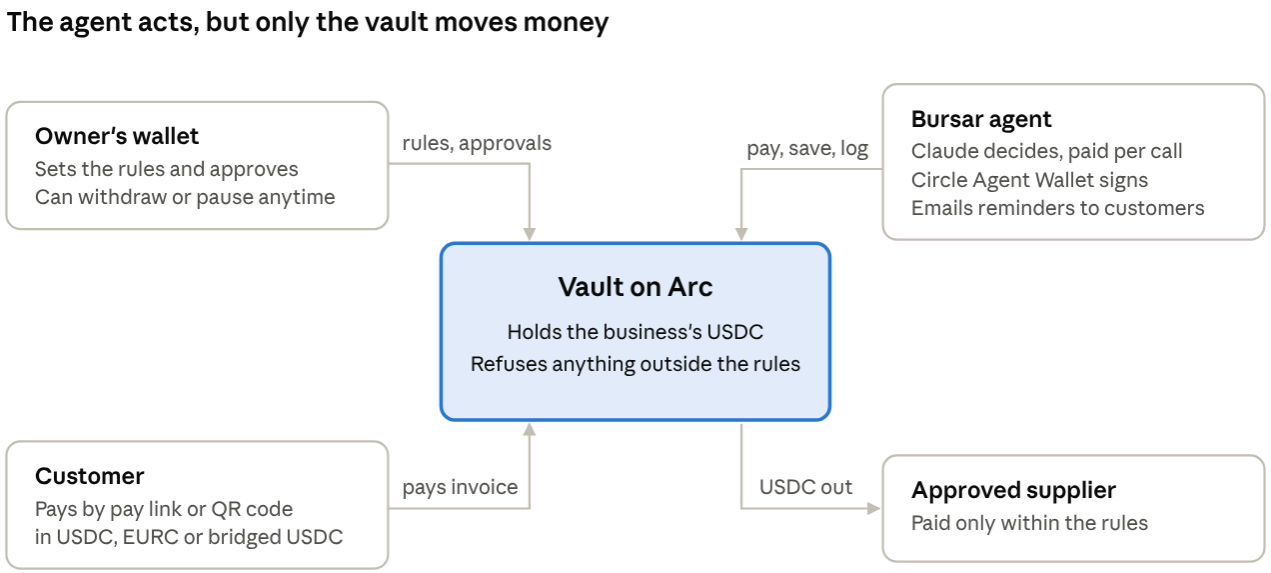
Each of those steps can go wrong:

- **Late payers slip through** because nobody had time to chase them.
- **Payments land unmatched,** so the owner can't tell which invoice a transfer was for.
- **Bills get paid twice,** or paid to the wrong wallet after a lookalike address or a fake invoice.
- **There's no record of why** money moved, which makes the books hard to trust.

An AI agent could do this work. But handing an agent the keys to a business's money is a real risk. It can make a mistake, or someone can trick it. Bursar answers that by letting the agent work while a contract it can't override holds the money and the rules.

How Bursar works

Bursar splits the job in two: an agent that decides, and a vault that has the final say.



How Bursar fits together

- **The vault** is a smart contract on Arc that only the owner's wallet owns. It holds the business's USDC and checks every payment against the owner's rules before any money moves.
- **The agent** reads invoices, payments and bills, decides what to do, and asks the vault to act. If a request breaks a rule, the vault refuses it on-chain, whatever the agent was told.
- **The app** at bursar.myacxt.site is where the owner sets up, sends invoices and watches the agent's decisions, and where customers pay.

The vault: rules on-chain

Every business gets its own vault, created in one transaction from the Bursar factory on Arc. The owner's wallet owns it. Only the owner can change its rules, approve the agent, or take money out.

Before the agent pays anything, the vault checks every rule below. If any check fails, the payment is refused on-chain.

Rule	What it does
Approved suppliers	The agent can only pay wallets the owner has approved
Most the agent may pay at once	The biggest single bill the agent can pay alone. Bigger bills wait for the owner's approval
Cash to always keep	A floor the agent never spends below
Category budgets	A spending limit per category, reset every 30 days
Reserve	Money set aside. The agent can add to it, but only the owner can release it
No double payments	The same bill reference can never be paid twice
Pause	The owner can stop the agent instantly

Whatever the rules say, the owner can always **withdraw** any amount, reserve included, and can remove or replace the agent at any time.

Savings. Each business can also have a savings vault with three kinds of pots:

- **Flex:** take money out anytime.
- **Locked:** can't be taken out before its unlock date. The agent can only fill it if the owner allows that for the pot.
- **Target:** a goal amount by a date, and money can come out anytime.

The agent can move money into savings when the business gets paid, but it can never take savings out. Only the owner can.

The agent

The agent runs the business by itself on a timer, every 6 hours by default, plus once after every restart. On each run it looks at the whole business:

1. **Matches payments** that arrived in the vault to the right invoice, flags overpayments, and emails the customer a "Payment received" note.
2. **Chases late payers** by email: a heads-up before the due date, firmer reminders once late, and an instalment offer when it helps. After 3 reminders it stops and hands the customer to the owner.
3. **Pays bills** that are due, inside the rules. Anything over a limit, to a new wallet, or that looks wrong goes to the owner to approve.

4. **Keeps a reserve** of spare cash and **saves** a share of each customer payment into savings pots.
5. **Writes every decision down** with its reason.

Its tools. The AI never touches money directly. It can only call these tools, and each one checks the facts in code before acting:

Tool	What it does
match_payment	Links a payment to an invoice, using only real transfers read from the chain
chase_customer	Sends a reminder email, at most 3 and no more than one a day
offer_instalments	Splits a late invoice into parts, only if the owner switched it on
pay_bill	Pays a supplier after duplicate, wallet and vault-rule checks pass
hold_bill	Waits on a bill, never past its due date unless cash forces it
request_approval	Hands a payment to the owner to decide
move_to_reserve	Sets spare cash aside
note	Writes an observation to the log, and nothing moves

Even if the AI says "pay", the code and then the contract decide.

How it thinks and signs. The agent's decisions come from Claude. It pays for each AI call in USDC from its own wallet through BlockRun on Circle's Agent Marketplace, with a daily spending cap of 0.25 USDC. It signs on Arc with a Circle Agent Wallet, so no private key sits on the server. One agent wallet serves every business, and it never holds a business's money.

Walkthrough: from sign-up to first payment

A business can go from nothing to its first paid invoice in a few minutes at bursar.myacxt.site. The screenshots below come from Bursar's test mode with a sample business, Ada Fabrics.

1. Pick a wallet

The owner's wallet will own the vault. There are three ways to set one up:

- **Connect a browser wallet** such as MetaMask, Rabby or Coinbase Wallet.
- **Create a new wallet on the site.** Bursar makes a real wallet with a 12-word recovery phrase and a Copy phrase button, and the same phrase works in MetaMask later.
- **Import a wallet** with its recovery phrase or private key.

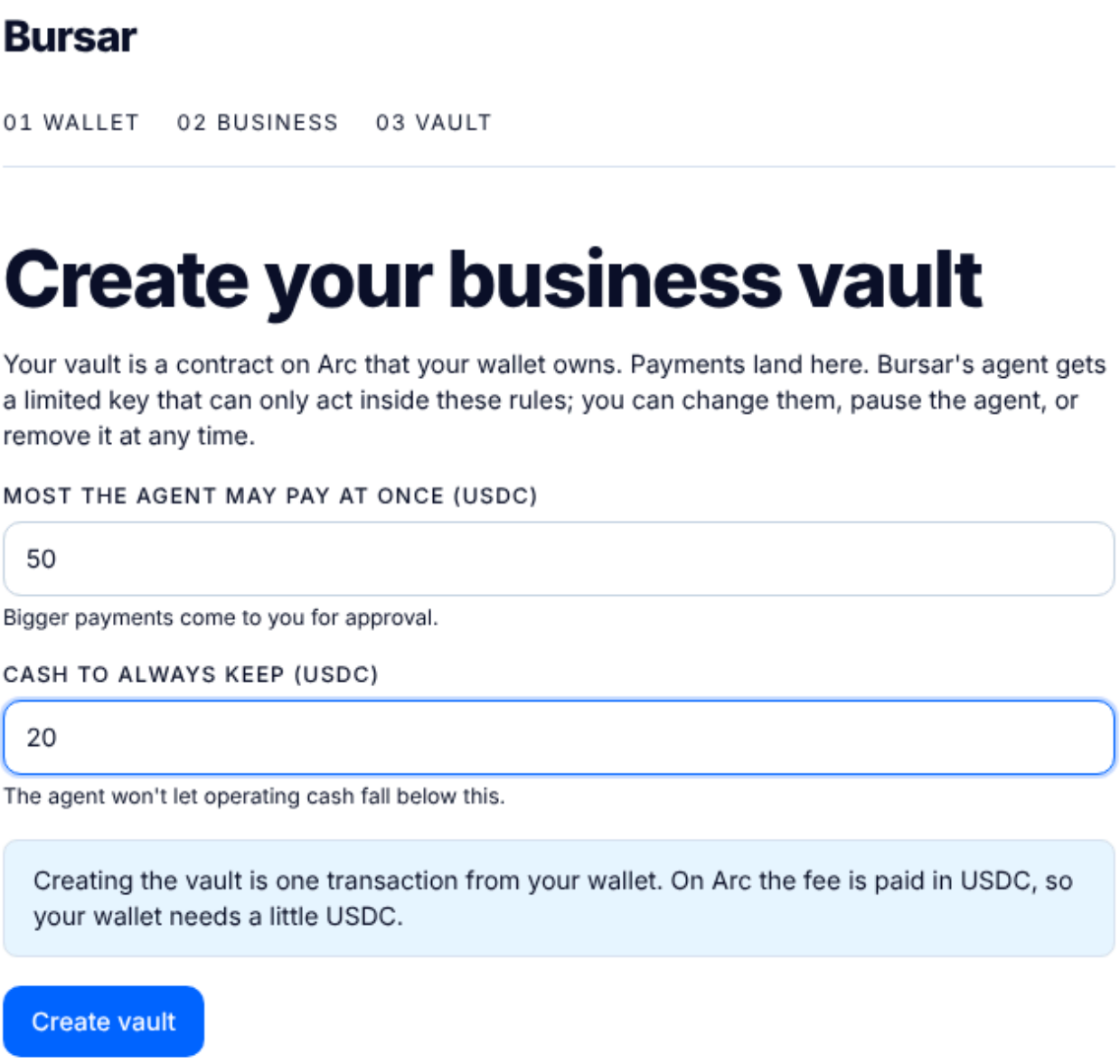
A wallet made or imported on the site is locked with a password, and its key stays encrypted in the owner's browser. It's never sent to Bursar. Before every on-chain action, Bursar shows its own Confirm transaction box with the fee and balance.

2. Name the business

The owner enters the business name, which customers see on invoices and reminders, plus an optional email for approval requests.

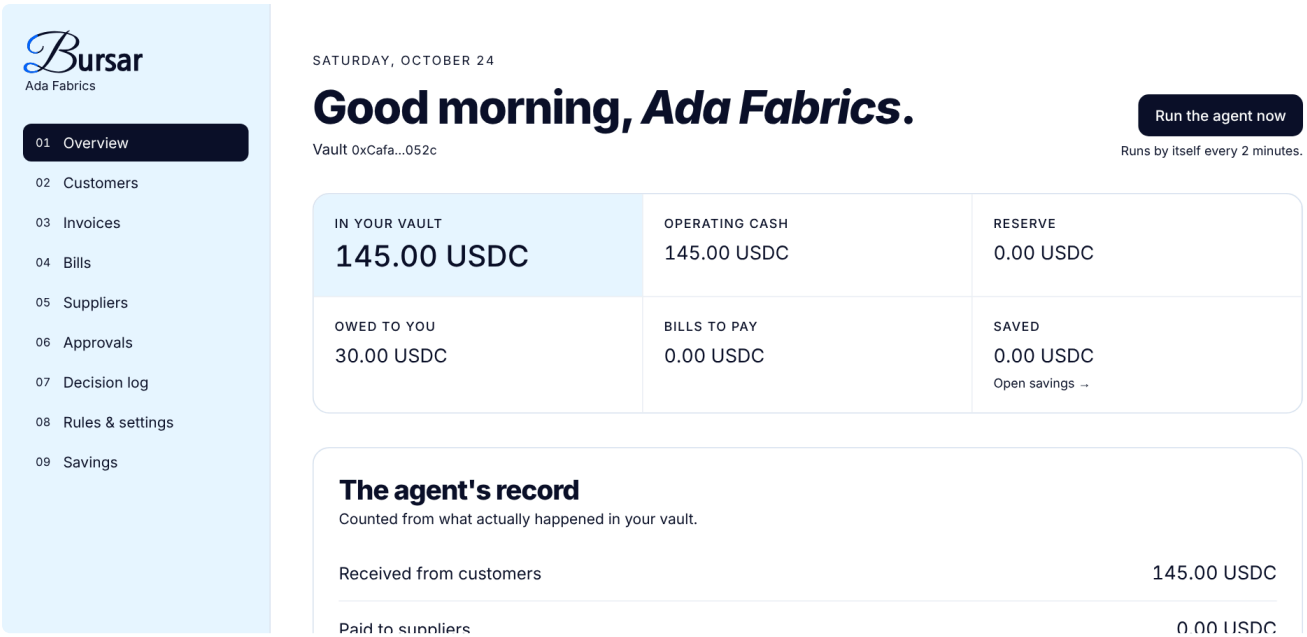
3. Create the vault

The owner sets two numbers: the most the agent may pay at once, and the cash it must always keep. Creating the vault is one transaction with a small one-off fee in USDC.



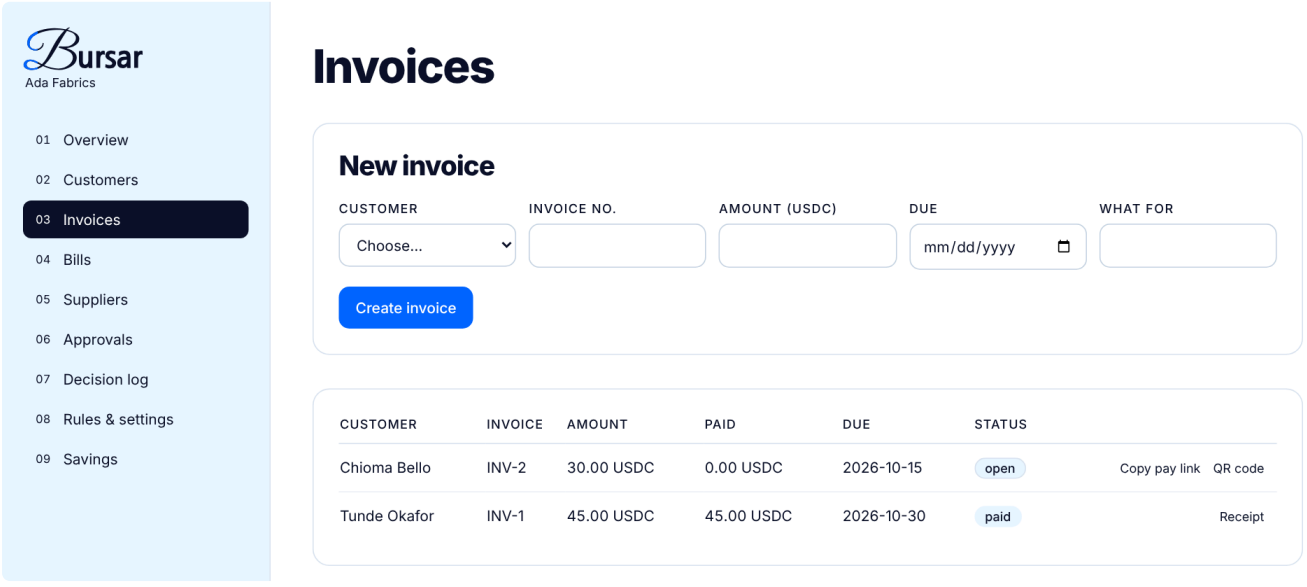
4. Watch the overview

The overview shows what's in the vault, operating cash, the reserve, what customers owe, bills to pay and savings. Below it, "The agent's record" counts what the agent actually did.



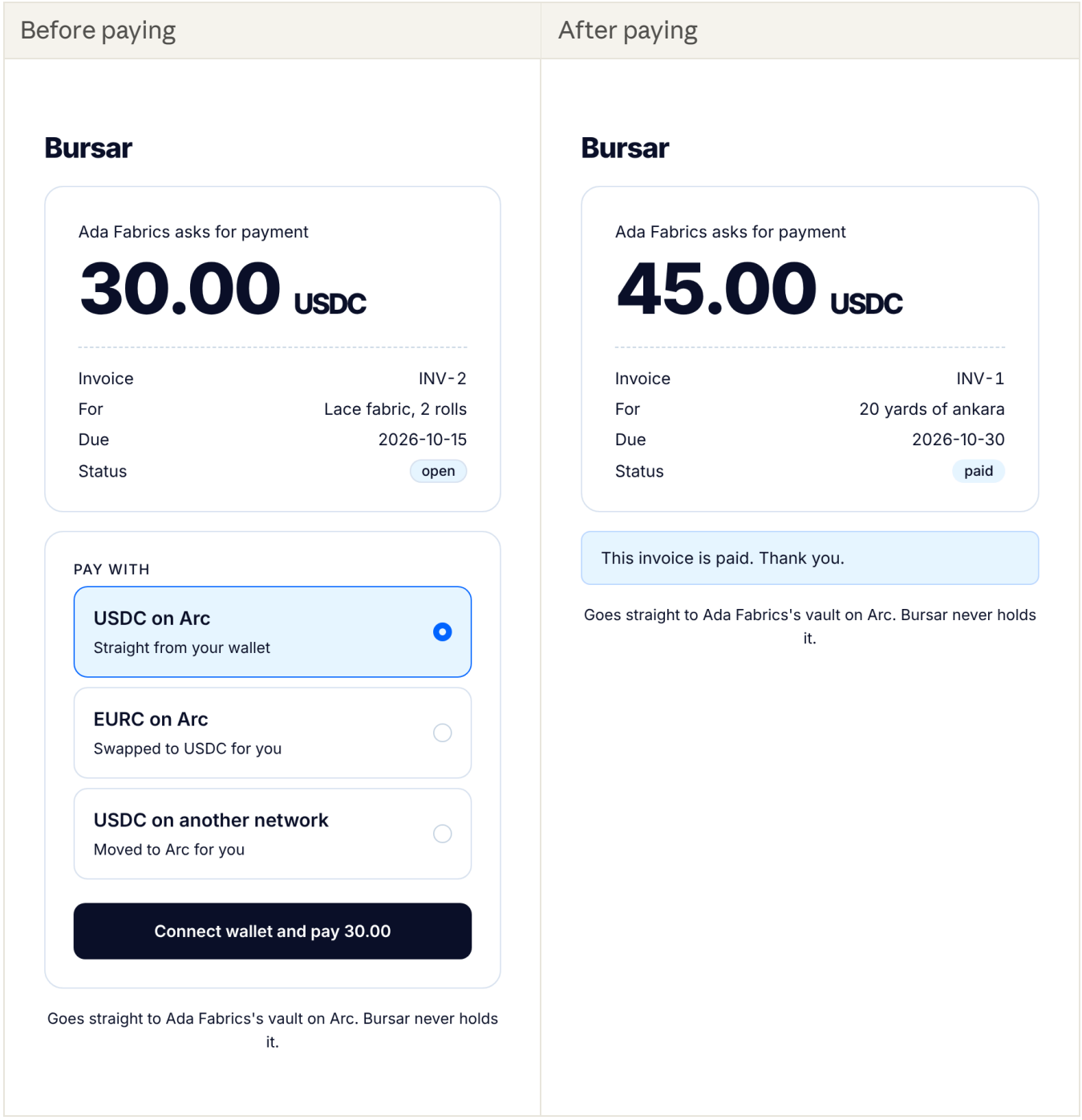
5. Add customers and send invoices

The owner adds a customer with their email and, optionally, their wallet, which is checked against sanctions lists straight away. Each invoice gets a pay link and a QR code to send in any chat.



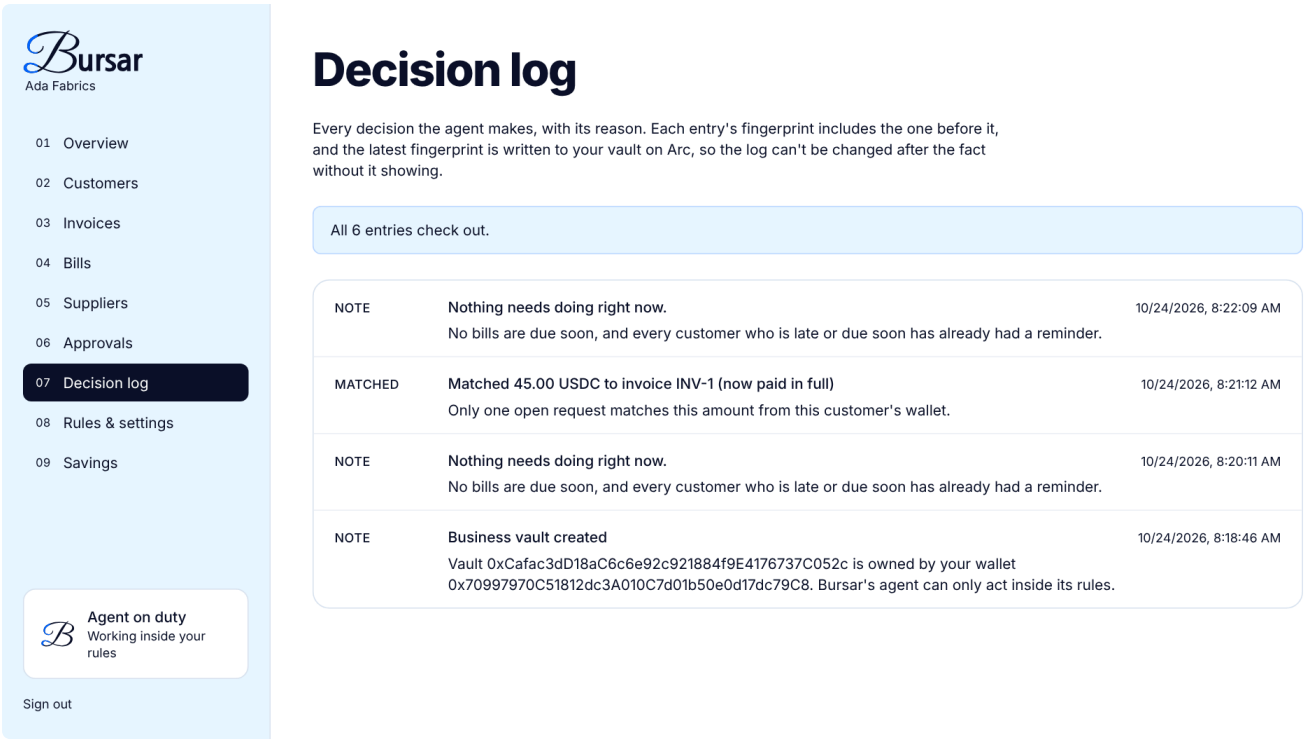
6. The customer pays

The customer opens the pay link on their phone and pays in USDC on Arc, EURC on Arc, or USDC from another network. The money goes straight to the business's vault, and Bursar never holds it.



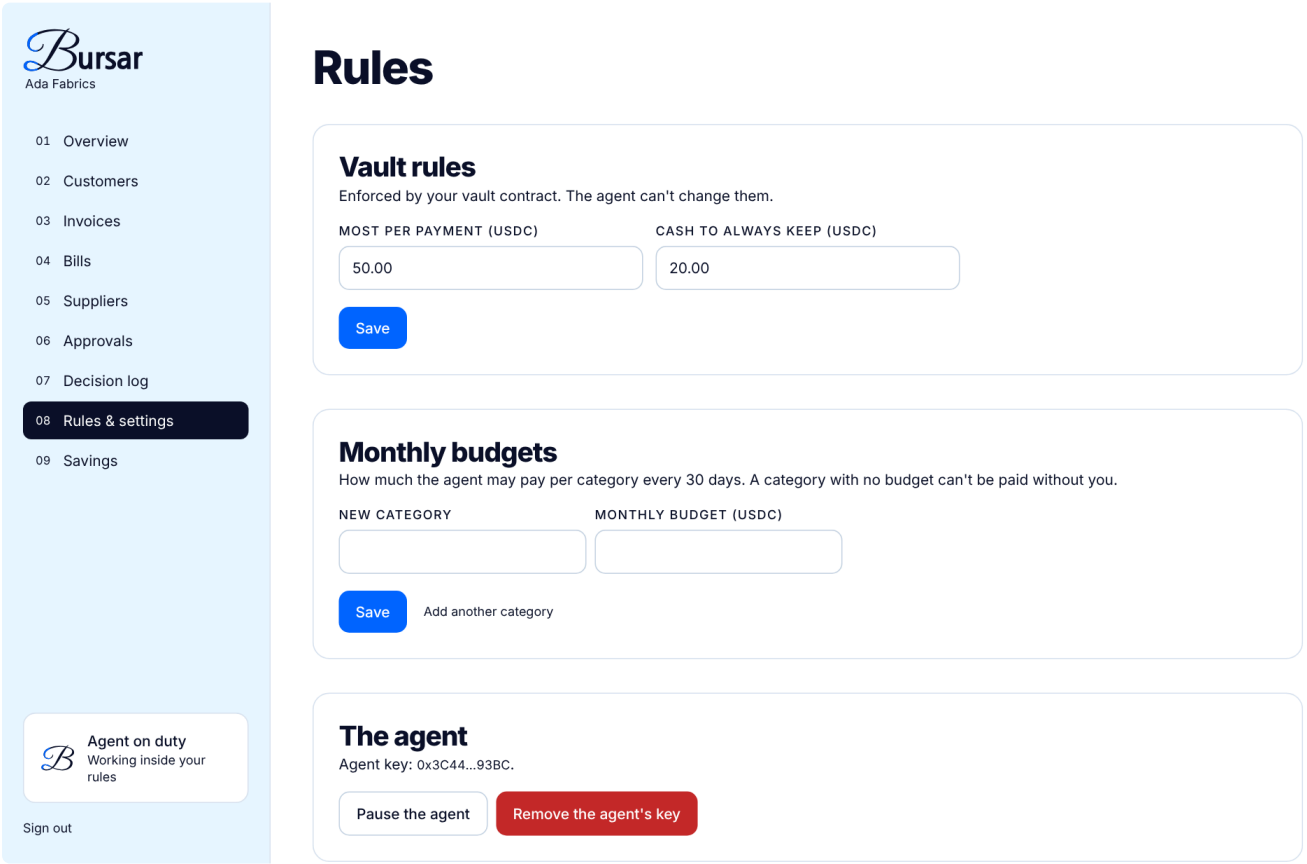
7. The agent takes over

On its next run, the agent matches the payment to the invoice and emails the customer a receipt. It also reminds anyone who's late. Every step goes into the decision log with its reason.



8. Stay in control

On Rules & settings, the owner can change limits, set monthly budgets per category, approve suppliers, pause the agent or remove its key. Every change goes to the vault in one approval. The owner can withdraw money at any time.



Safety

Bursar assumes the agent will sometimes be wrong or be tricked. Each likely mistake has a check that stops it:

What can go wrong	What stops it
The agent pays more than it should	The per-payment limit and monthly budgets, enforced by the vault
The agent drains the account	A cash floor the vault won't spend below; only the owner can use the reserve
The agent pays a scammer's wallet	Payments only to approved wallets, plus sanctions, blocklist, lookalike-address and wallet-change checks
The same bill is paid twice	The vault records each bill reference and refuses a second payment
A fake or duplicate bill	Duplicate invoice number and same-amount-same-supplier checks; mismatches go to the owner
A payment that never arrived is marked paid	Payments are only matched from real USDC transfers read from the chain
A hidden instruction in a customer's name	Messages can't contain wallet addresses or links; the app adds the real pay link itself
The record is changed later	A hash-chained decision log, anchored in the vault
The agent itself goes wrong	The owner can pause it, remove its key, or withdraw everything at any time

Wallet checks. Every customer and supplier wallet is checked against the US Treasury (OFAC) sanctions list and Circle's USDC blocklist when it's added, and every wallet is re-checked daily.

A log that can't be quietly edited. Each decision-log entry is hashed together with the one before it, and the agent writes the latest hash to the vault on Arc. If anyone changes a past entry, the chain stops matching the hash on-chain, and the log page shows it.

Built on Circle and Arc

Bursar runs on Arc mainnet, where fees are paid in USDC, so a business only ever needs one currency. Each Circle tool does one part of the job:

Tool	What it does in Bursar
USDC on Arc	All business money: vaults, the reserve, savings and the agent's own AI payments
Circle Agent Wallet	The agent signs every on-chain action through it, so no agent private key sits on the server
Circle Agent Marketplace (x402)	The agent pays BlockRun for each Claude call in USDC, with a cap of 0.10 USDC per call and 0.25 USDC per day
Circle App Kit	Customers can pay in EURC (swapped to USDC) or bridge USDC from another network with CCTP, straight into the vault
USDC blocklist	Every wallet is checked against Circle's blocklist before money moves

The agent pays for its own thinking, so neither Bursar nor the business needs an AI subscription, a card or an API key.

Traction

Bursar is live on Arc mainnet with real money. Five businesses have been onboarded so far, and the first real customer payment has already landed in a vault and been matched by the agent.

How a business is onboarded:

1. I walk the owner through setting up a wallet, online or in person, including what USDC and Arc are.
2. The owner creates their own vault on Arc mainnet and sets their rules.
3. They add their first customer and send their first invoice.
4. From there, the agent runs the money work on its own.

What's next. I'm onboarding more businesses through marketing, and I'll update this whitepaper as that grows.

Contracts, links and limits

Both factories are verified on the Arc explorer, and the code is open source under the MIT licence.

Item	Where
Live app	bursar.myacxt.site
BursarFactory	0x1d872C07C1935fe5E3a28a9df0690C2caf7B5709
BursarSavingsFactory	0xe7395461185d3d8964dDaA29B31b963Cfb96CB1c
Bursar code	github.com/Mhiah/Bursar-
Vault kit for other apps	github.com/Mhiah/bursar-vault
Demo video	x.com/myacxt
Contact	@myacxt on X

Limits

- Bursar isn't a bank. Money sits in a contract the owner's wallet owns.
- Lost wallet keys can't be recovered by anyone, including Bursar.
- The vault holds USDC only. Customers can pay in EURC or from other networks, but it arrives as USDC on Arc.
- Wallet checks lower risk; they don't remove it. New wallets always go to the owner first.
- Bursar keeps a clear record, but tax and filing are still the owner's job.

Bursar is a product built for the Tameion Agents Hackathon (Canteen x Circle x Arc). Your money stays in a vault only your wallet owns, and the agent can only act within the rules you set.